



Logistics Delivery Risk Intelligence

Delivery Delay Prediction & Operational
Intervention for a Multi-Modal Logistics
Network

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Logistics Delivery Risk Intelligence

1 Assignment Overview

This is the second scenario-based assignment for the Applied Machine Learning module, designed to be completed in a single focused 150-minute sitting — the kind of time-boxed analytics sprint a logistics operations analyst is actually given when leadership asks for an answer before the next planning meeting.

You will apply the same end-to-end applied ML lifecycle as Assignment 1 — data validation, preprocessing, feature engineering, regression, classification with ensembles, K-Means clustering, and PCA — to a new domain and a new failure mode: predicting which shipments are at risk of arriving late, early enough that a planner can still act.

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2 Real-World Business Scenario

Her brief:

The Brief

"We already know a shipment was late once it's late — that's not intelligence, that's a report card. I need to know which shipments are at risk while they're still in the network, so my planners can re-route, expedite, or call the customer ahead of time. Show me the evidence behind every recommendation."

— Head of Operations, RouteWise Logistics

Your job is to translate this brief into two predictive questions and one exploratory question:

1. Forecasting — How many hours of delay should we expect for a given shipment? (regression)
2. Decisioning — Should this shipment be flagged as delay-risk for planner intervention? (classification)
3. Understanding — What recurring operating patterns exist across RouteWise's shipment network, and which ones carry the most risk? (clustering & dimensionality reduction)

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3 Dataset Description

You are provided with logistics_delivery_delay.csv – shipment-level records drawn from RouteWise's dispatch and tracking systems.

3.1 At a Glance

Property	Detail
Rows	720 shipment records
Columns	17 (14 predictors/context + 1 post-outcome field + 2 targets)
Time Coverage	1 April – 30 June 2026, sampled every 3 hours
Origin hubs	North, South, East, West (180 shipments each)
Shipping modes	Road, Air, Rail (240 shipments each)
Destination types	Metro, Industrial, Rural, Tier-2 City (180 each)

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3 Dataset Description

3.1 At a Glance

Property	Detail
Carriers	Carrier-A, Carrier-B, Carrier-C, Carrier-D (180 each)
delay_required positive rate	≈ 69.3% – imbalanced toward the delayed class

3.2 Column Reference

Column	Type	Description
destination_type	categorical	Metro / Industrial / Rural / Tier-2 City
shipping_mode	categorical	Road / Air / Rail
carrier	categorical	Carrier-A _ Carrier-D
distance_km	numeric	Route distance

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3 Dataset Description

3.2 Column Reference

Column	Detail	Description
package_weight_kg	numeric	Package weight
weather_condition	categorical	Clear / Cloudy / Rain / Storm
traffic_levelgion	categorical	Low / Medium / High
carrier_rating	numeric	Carrier performance rating
warehouse_processing_hours	numeric	Time spent in warehouse processing
dispatch_delay_hours	numeric	Delay before dispatch
handling_time_hours	numeric	Shipment handling duration

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3 Dataset Description

3.2 Column Reference

Column	Type	Description
expected_delivery_hours	numeric (%)	Average memory utilization
actual_delivery_hours	numeric-POST-OUTCOME	Network throughput
delivery_delay_hours	numeric-Target 1	Realized delay beyond expected delivery (regression target)
delay_required	binary-target 2	1 = delay-risk classification, 0 = not flagged (classification target)

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3 Dataset Description

Known Data Quality Notes (do not fix blindly – investigate first)

- Small numbers of missing values exist in package_weight_kg, weather_condition, traffic_level, carrier_rating, and warehouse_processing_hours.
- The dataset is supplied as-is from RouteWise's tracking system; do not fabricate, delete, or manually alter any record. Document every preprocessing decision.
- actual_delivery_hours, delivery_delay_hours, and delay_required are all determined only after a shipment has actually been delivered. None of the three may ever be used as a predictor of the others, and none may enter the unsupervised feature set.

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4 Learning Outcomes

By completing this assignment you will demonstrate the ability to:

- Audit and validate a real, imperfect operational dataset before modelling, including cross-category comparisons.
- Engineer domain-driven features for a logistics context without introducing target or post-outcome leakage.
- Build and evaluate a regression model for delay-magnitude prediction under a strict chronological split.
- Build, compare, and select between a Decision Tree and a Random Forest classifier under class imbalance.
- Apply K-Means clustering and PCA to discover recurring shipment operating patterns.
- Run a structured validation pass on your own pipeline and communicate results as an operational recommendation.

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5 Ground Rules

1. Use the dataset exactly as supplied. Do not fabricate, duplicate, or hand-edit records.
2. Data-leakage rule: `actual_delivery_hours`, `delivery_delay_hours`, and `delay_required` must never be used as predictor variables.
3. Post-outcome information — anything only known after the shipment is delivered — must not be used to construct features.
4. Use a chronological (time-aware) train/evaluation split for the regression and classification tasks — do not shuffle randomly across time.
5. Fit all data-dependent preprocessing (scalers, encoders, imputers) on the training partition only, then apply to the evaluation partition.
6. Exclude `delivery_delay_hours`, `delay_required`, `actual_delivery_hours`, and every other post-outcome field from the unsupervised feature set.
7. Report the actual evaluation metrics produced by your implementation — never invent or estimate a score.
8. For classification, do not rely on Accuracy alone; interpret Precision, Recall, F1-score, and the Confusion Matrix together.

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6 Assignment Tasks

Total: 40 marks across 8 tasks, aligned to the Data Pre-processing, Feature Engineering, Regression, Sessionification, and Unsupervised Learning sessions of this module.

Task 1 – Understand and Validate the Shipment Data

4 Marks

- Load the CSV and report the record count, column count, and data types.
- Report the time coverage and the distribution of every categorical column (origin_hub, destination_type shipping_mode, carrier, weather_condition, traffic_level).
- Report the distribution of both targets – flag that delay_required is imbalanced (~69% positive).
- Identify and quantify all missing values and where they occur.
- Inspect numerical variables for unusual or extreme observations.
- Compare observed delay behavior across shipping mode, traffic level, weather condition, destination type, and carrier – where is delay concentrated?
- Verify that the target columns are separated from predictor columns before modeling.



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6 Assignment Tasks

Task 2 – Prepare the Data

5 Marks

- Convert shipment_timestamp into useful time-based predictors where justified (e.g. hour of day, day of week).
- Encode the categorical variables (origin_hub, destination_type, shipping_mode, carrier, weather_condition, traffic_level) appropriately.
- Handle the supplied missing predictor values and document the chosen approach for each affected column.
- Apply scaling where required by the selected algorithms, and explain where it is unnecessary.
- Inspect outliers using an appropriate method (IQR or Z-score) and apply treatment only when justified by evidence.
- Ensure all data-dependent preprocessing is fit on training data only.

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6 Assignment Tasks

Task 3 – Engineer and Select Features

5 Marks

- Create at least one business-relevant feature representing shipment complexity, route pressure, or operational delay pressure — e.g. a composite of dispatch delay + warehouse processing + handling time, or distance relative to expected delivery hours.
- Explain why the engineered feature is useful to the logistics problem — what would a planner read off it?
- Use an appropriate feature-selection or elimination method (correlation, RFE, or tree-based importance) to identify weak or redundant predictors.
- Document which features were retained and which were removed, and why.
- Demonstrate explicitly that no target or post-outcome field has entered the predictor set

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6 Assignment Tasks

Task 4 – Predict Delivery Delay (Regression)

6 Marks

- **Target:** delivery_delay_hours.
- Use only pre-outcome predictors – no actual_delivery_hours or other post-outcome fields.
- Create a chronological train/evaluation split respecting shipment_timestamp order.
- Build and evaluate a Linear Regression model.
- Report MAE and RMSE on the evaluation period.
- Create an actual-vs-predicted delay visualization.
- Briefly interpret the size and pattern of the prediction errors – where does the model struggle?

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6 Assignment Tasks

Task 5 – Predict Shipment Delay Risk (Classification)

6 Marks

- **Target:** delay_required.
- Build a Decision Tree classifier using pre-outcome predictors only.
- Build a Random Forest classifier using the same evaluation period.
- Report Accuracy, Precision, Recall, F1-score, and the Confusion Matrix for both models.
- Explain why Precision and Recall matter differently for a logistics intervention decision – a missed delay-risk shipment (false negative) reaches the customer late with no warning, while a false alarm (false positive) costs planner time on a shipment that would have arrived fine.
- Compare the two models using your actual evaluation results and select the more useful one for RouteWise's planning workflow.

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6 Assignment Tasks

Task 6 – Discover Shipment Operating Patterns (Clustering)

5 Marks

- Create one documented unsupervised feature set containing only pre-outcome shipment characteristics – all target and post-outcome fields excluded.
- Use K-Means clustering to identify recurring shipment operating patterns.
- Evaluate a reasonable range of candidate cluster counts (e.g. Elbow method and/or Silhouette score) and justify your selected K.
- Describe the operational characteristics of each resulting cluster in plain language.
- Explain which cluster(s) may deserve closer operational attention, and why.

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6 Assignment Tasks

Task 7 — PCA and Delivery-Risk Visualization

4 Marks

- Apply PCA to the same prepared unsupervised feature set and reduce it to two components.
- Create a 2D PCA visualization.
- Create at least one visualization comparing delay behavior across an important operational category (e.g. shipping mode, traffic level, or weather).
- Create a model/feature-insight visualization (e.g. feature importance from the Random Forest).
- Create a K-Means cluster visualization.
- Use clear titles, axis labels, legends, and units throughout.

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6 Assignment Tasks

Task 8 — Validation, Testing and Operational Recommendation

5 Marks

- Perform a normal-case data and pipeline validation check on a small held-out sample.
- Verify that missing values are handled before model fitting.
- Verify that target and post-outcome fields are excluded from every predictor set, including the unsupervised feature set.
- Verify that the chronological evaluation split is respected end-to-end.
- Verify that the regression metrics and the classification confusion matrix are genuinely generated from the evaluation data, not the training data.
- Summarize the most important operational findings and recommend how RouteWise should prioritize intervention.
- State at least two limitations of the analysis.

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7 Expected Outcomes

- Produce a clean, documented machine-learning workflow using the supplied shipment dataset.
- Predict delivery-delay magnitude using Linear Regression and report actual MAE and RMSE.
- Predict delay risk using Decision Tree and Random Forest and compare their evaluation metrics.
- Identify recurring shipment operating patterns using K-Means.
- Use PCA and clear visualizations to communicate the structure of the data.
- Demonstrate that target leakage is avoided and the evaluation process is valid.
- Provide an evidence-based recommendation for proactive logistics intervention.

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8 Deliverables

Deliverable	Must Contain
logistics_delivery_delay_analysis.ipynb	Complete analysis: preprocessing, feature engineering & selection, regression, classification, clustering, PCA, validation checks, visualizations, evaluation results, and recommendation.
README.md	Business problem, dataset description, preprocessing decisions, model results, clustering findings, limitations, and final recommendation.
Results section	Regression metrics, classification comparison, confusion matrices, and key visualizations, either included directly in the notebook or exported to the outputs/ directory.

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9 Evaluation Rubric

Task	Must Contain	Marks
1	Data Understanding & Validation	4
2	Data Preparation	5
3	Feature Engineering & Selection	5
4	Regression (Delivery Delay)	6
5	Classification (Delay Risk – Decision Tree vs Random Forest)	6
6	Shipment Operating Pattern Discovery (K-Means)	5
7	PCA & Delivery-Risk Visualization	4
8	Validation, Testing & Recommendation	5
	Total	40
Bonus	Hierarchical/DBSCAN + Intervention Framework (optional)	Bonus

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10 Bonus Challenge (Optional)

This section does not count toward the core 40 marks, but strong submissions are noted for the program's Capstone Showcase.

1. Apply Hierarchical Clustering to the same unsupervised feature set and interpret the dendrogram.
2. Apply DBSCAN and report the resulting clusters and noise observations. If the result is not useful, explain why rather than forcing an interpretation.
3. Design a simple low/medium/high intervention-priority framework using the completed model outputs.
4. Recommend one additional real-world data source that could improve a production delivery-risk system.

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11 Submission Guidelines

- Submit the notebook and README as specified in Section 8 through the LMS assignment portal.
- Ensure the notebook runs top-to-bottom without errors before submission — a broken kernel state will not be re-run for grading.
- Cite any external library documentation or reference material you consulted.
- Late submissions follow the program's standard late-day policy; contact your mentor in advance for extensions.

A Note From the ML & Analytics Guild Lead

This one is time-boxed for a reason — 150 minutes is close to what a working analyst actually gets. Prioritize a working, honest pipeline over a perfect one: a Task 8 validation pass that catches your own leakage is worth more than a fancier model you didn't have time to check.